

Day 24 Hybrid Notes

Lines, distances, circles, and Algebra 2 connections | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets

- I can choose slope, distance, midpoint, or circle equations to solve a problem.
- I can use coordinate evidence to classify figures.
- I can connect circle equations to transformations and conics.

Part 1: Cloze Notes - Essential Ideas

- Slope measures vertical change divided by horizontal _____.
- Parallel lines have equal _____.
- Perpendicular lines have slopes that are negative _____.
- The distance formula comes from the _____.
- The midpoint formula averages the x-values and the _____.
- A circle equation in center-radius form is $(x-h)^2+(y-k)^2=$ _____.
- A circle is one example of a conic _____.

Part 2: Outline Notes - Big Picture

- A. Coordinate geometry turns diagrams into algebraic _____.
- B. Slope is useful for proving parallel or perpendicular _____.
- C. Distance is useful for proving congruent sides or finding _____.
- D. Transformations of equations prepare students for Algebra 2 _____.

Connection to Prior Math

- What algebra skill shows up today?

- What diagram or graph skill shows up today?

- What is one place this could appear outside math class?

Most Important Reminder

Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 24 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. Write the equation of a circle with center (3,-4) and radius 6.

Setup/work: _____

Answer: _____

2. Describe the transformation from $x^2+y^2=25$ to $(x-3)^2+(y+2)^2=25$.

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student says $(x+4)^2+(y-3)^2=25$ has center (4,-3).

What went wrong?

Correct idea:

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Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A quadrilateral has vertices (0,0), (5,0), (5,5), and (0,5).

Decide whether it is a square.

Setup: _____

Answer: _____

Sentence answer: _____